

Highlights

Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

Haoran Ma, Ningning Wang

- SC-FMA calibrates interventional utility into supervision weights.
- QP is strongest only on the controlled synthetic proxy benchmark.
- PRM800K real evidence is led by `w_struct`; Ridge is closest.
- Audit-prioritization evidence is offline locked-split context only.
- Failed GSM8K/HotpotQA and downstream routes are kept out of claims.